

Mark Lourenco Rojas

Software Developer
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Education

10.2023 / 10.2026

LaSalle College Vancouver

Vancouver, BC

Bachelor's Degree in Computer Games and Programming Skills

05.2026 / 08.2026

Centre for Digital Media

Vancouver, BC

CDM Practicum Program

03.2022 / 03.2023

Vancouver Institute of Media Arts

Vancouver, BC

Diploma in Game Art & Design

Expertise

Proficient in:

- Unreal Engine, Unity, Maya 3D
- Substance Painter & Designer
- Visual Studio
- Adobe Photoshop
- Github, Perforce

Familiar with:

- Zbrush
- Marvelous Designer
- Android Studio
- Houdini

Skills

- C++, C#, Unreal Blueprints
- Data Structures and Algorithms, OOP, ECS, Debugging
- AI Behavior, Pathfinding
- Multi-threading, Parallel Programming
- Hard-surface modeling, modularization
- Unreal Niagara & Cascade VFX, Unity VFX
- Basic rigging & animation
- Quick prototyping, Game Sense & Design
- Collaboration, Effective Communication with Artists and Programmers

Languages

English - Fluent

Spanish - Fluent

Work Experience

12.2025

Unreal Engine Blueprint Instructor

Taught **Unreal Engine Blueprint** classes, teaching new students about **interaction systems**, several **platforming mechanics** (moving platforms, character movement), **animation**, and **level design** while also helping them learn the **Unreal User Interface**.

Projects

05.2026 / 08.2026

Magic Trails | Educational AR Mobile Experience

Centre for Digital Media Practicum

Gameplay Programmer, Audio Direction/Programmer



- Engineered **AR gameplay mechanics** such as **plane-anchored placement**, **hand-gesture recognition**, and **directional audio beacons**, to ground each mission in the player's real environment.
- Implemented guided UI flows including tooltips, progress indicators, and completion overlays so players could complete every mission **without external tutorials**.
- Built a **Unity Audio Manager** supporting layered adaptive music with **sample-accurate stem toggling**, spatialized 3D and 2D SFX, per-channel volume control, and cross-scene music persistence, powering all audio for the app's missions and menus.
- **Led audio design direction** including music, SFX, and dialogue for all audio to fit into the desired world of the app.
- **Worked with a multi-disciplinary team for a real client**, building a full demo of their vision, to translate their concept into a playable product they could evaluate.

01.2023 / 05.2023

Rute: The Spirit Dish | 3D Stealth-and-Puzzle Adventure

The Rookies

Lead Programmer, Systems Designer, Gameplay Programmer, Gameplay Designer



- **Contributed to game concept, puzzle design, and gameplay flow** during pre-production, **guiding scope decisions with engineering constraints** to keep the project **shippable within a tight timeline**.
- Transitioned into a gameplay programmer role and shipped core **Unreal Engine** systems for a 3D stealth-and-puzzle adventure.
- Built **player movement and camera control systems** for third-person exploration through a forest.
- Implemented **dialogue and puzzle logic** to drive the game's narrative pacing and moment-to-moment challenges.
- Developed **stealth mechanics** and enemy AI governing how enemies detected, tracked, and reacted to the player.

02.2026

Masquerade Murder Night | 2D Side-Scroller Social-Deduction Party Game

Global Game Jam 2026

Gameplay Designer, Level Designer, Gameplay Programmer



- **Collaborated with other designers to define the game concept**, from **core genre and mechanics to gameflow and level layout**, establishing the foundation the team built on.
- Built core **AI pathfinding** for NPCs using a **node system, with clumping, popular-area gathering, and unpredictable behavior**, to make NPCs behave like real party guests and stay hard to distinguish from real players.
- Implemented gameplay mechanics within a **networked environment**, to keep behavior **synced and consistent across different players' machines**.
- **Created and integrated 2D sprite animations** using assets from 2D artists, using animators to drive animation flow and state switching, giving characters fluid, responsive movement.

Hobbies

- Game Jams
- Videogames (RPGs, CRPGs, Auto Chess...)
- TTRPGs (DnD 5e, Alien RPG...)
- Archery
- Music
- Gym

12.2022 / 02.2023

Pawl's Impossible Journey | 3D Platform Adventure Game

Vancouver Institute of Media Arts

Gameplay Programmer



- Built **rapid gameplay prototypes** during pre-production to help the art team visualize interactive mechanics and align environment design with playable scope.
- Developed **player movement and platforming mechanics** driving third-person traversal through the game's woods and forest environments.
- Implemented **checkpoint, collectible, obstacles, and level-transition systems** that structured the game's progression flow.
- Built **enemy actors, a damage system, and a dialogue system** to drive combat encounters and narrative moments.
- Supported the art team by **creating VFX and procedural textures** used across environments and gameplay events.

09.2025 / 03.2026

The Fool | 2D Deck-Building Roguelite Turn-Based RPG

LaSalle College Vancouver Production

Lead Programmer, Systems Designer, Gameplay Programmer, Gameplay Designer



- Built a **scalable card system** supporting unique icons, pricing, and active/passive effects per card, and **designed a partial card set** with balanced strengths and weaknesses across categories, enabling varied strategy against different enemies and scenarios.
- Developed a **turn-order system** driven by player and enemy speed stats, to govern the flow and pacing of combat.
- Constructed a **scalable enemy system** supporting unique skills and stats per enemy, enabling **varied encounter design without duplicating code**.

01.2025 / 01.2025

Asphyxia | 2D Resource-Management Choice-Based Game

Global Game Jam 2025

Gameplay Programmer, Audio Programmer



- **Co-developed the game concept** during pre-production, contributing to the **moral-choice trading mechanic** where players weigh profit against helping struggling visitors.
- Built the main menu and scene-transition flow to onboard players into the game loop.
- Implemented **player interaction and 2D movement mechanics**.
- Developed a **currency management system** to track money along with adjusting the value.
- Integrated **music and SFX to reinforce the emotional weight** of each trade decision along with the exhilarating passing of time.

